

SCE 301

Communication Systems



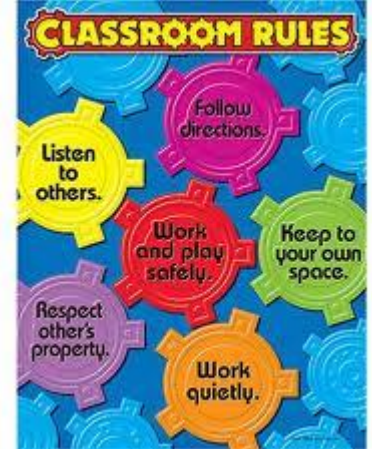
Class Rules

- It will be a little bit tuff class , however,
 - Enjoy what you do
- Do not worry about the exam as long as :
 - You are attending
 - Read the textbook and the provided materials
 - Aware of everything I mention in class
 - Done with assignments

Listen to the
teacher and to
others.



Class Rules



- I am not here to punish you
- You are the best and *I want you to continue the best*
- Trust yourself and do your best
- I want you to learn and compete with others working on the same field
- I want you to be confident when you speak with others

Class Rules

- You can bring anything to drink but
 - NO FOOD PLEASE ☺
- When you come in, **DO NOT** knock on the door
- When you want to leave, do not tell me Just leave but you will be counted as absent if you don't come back
- **Attendance is very important**
- **Assignments** must be delivered on time

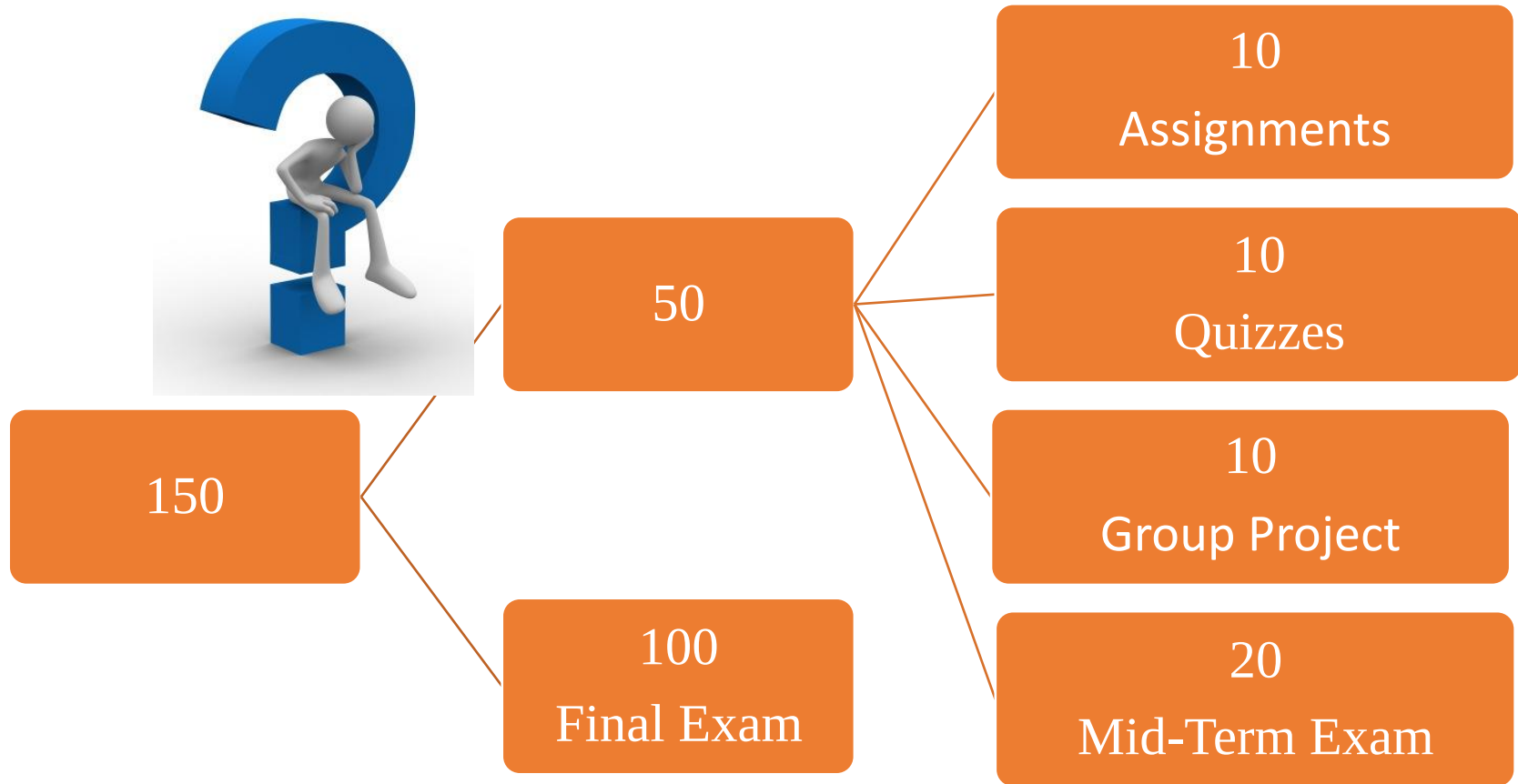
Course Contents

Topics	Week
Introduction: Communication Model, Data Communications, Networking	1,2
Communication Protocols: Protocols (characteristics and functions) and Protocol Architecture (OSI and definition of TCP/IP models)	3,4
Data Transmission: Data Transmission (concepts and terminology), Analog and Digital Data Transmission, Transform Representation, Transmission Impairments – Nyquist formula and Shannon's Capacity	5,6
Transmission Medium: Guided & Unguided Medium, Satellite Communication, Point to Point Link	7
Signal Encoding Techniques: Digital Signals Digital Data – Analog Data Digital Signals, Digital Data Analog Signal	9,10
Error Detection and Correction: Error Detection, CRC, Parity Check, Flow Control	11,12
Data Link Control Protocols: Flow Control (stop-and-wait and sliding window flow) and Error Control, HDLC	13
Multiplexing: FDM and Synchronous TDM, Synchronous TDM, Statistical TDM, and ADSL	14
Advanced Topic	15

Course Contents (Modified)

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6	Error Detection and Correction: Error Detection, CRC, Parity Check, Flow Control	11
7	Multiplexing: FDM and Synchronous TDM, Synchronous TDM, Statistical TDM, and ADSL	12
8	Fog Computing: Wireless Sensor Networks, Internet of Things (IoT), Cloud Computing, Fog Computing Applications with the IoT	13,14
9	Review	15

Grading Policy



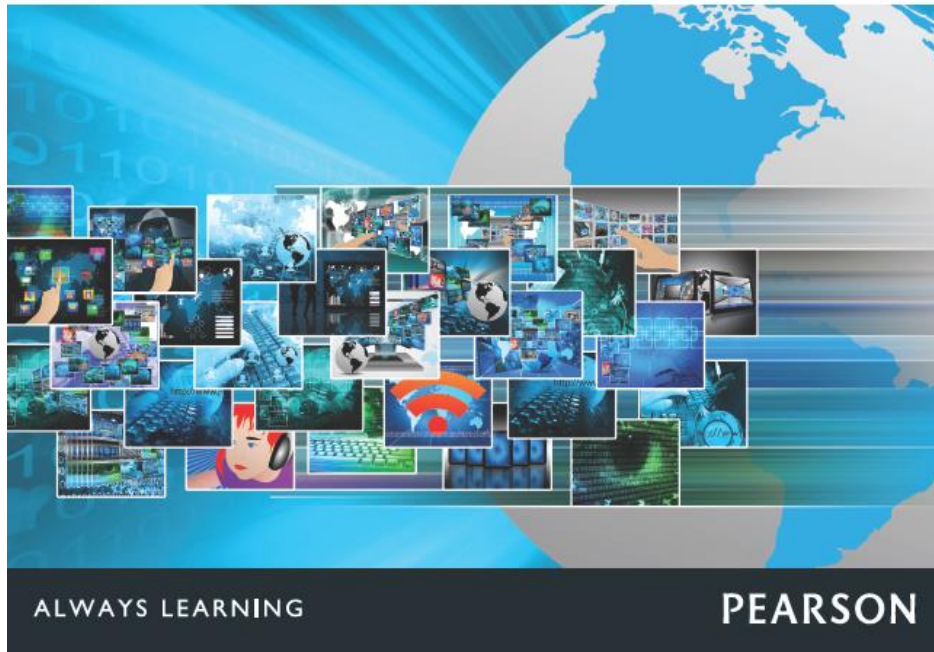
INTERNATIONAL
EDITION



Data and Computer Communications

TENTH EDITION

William Stallings



ALWAYS LEARNING

PEARSON

Behrouz A. Forouzan

Data Communications and Networking

Fourth Edition



CHAPTER 1

Data Communications, Data Networks, and the Internet

“Communication is almost as important to us as our reliance on air, water, food, and shelter. In today’s world, through the use of networks, we are connected like never before.”



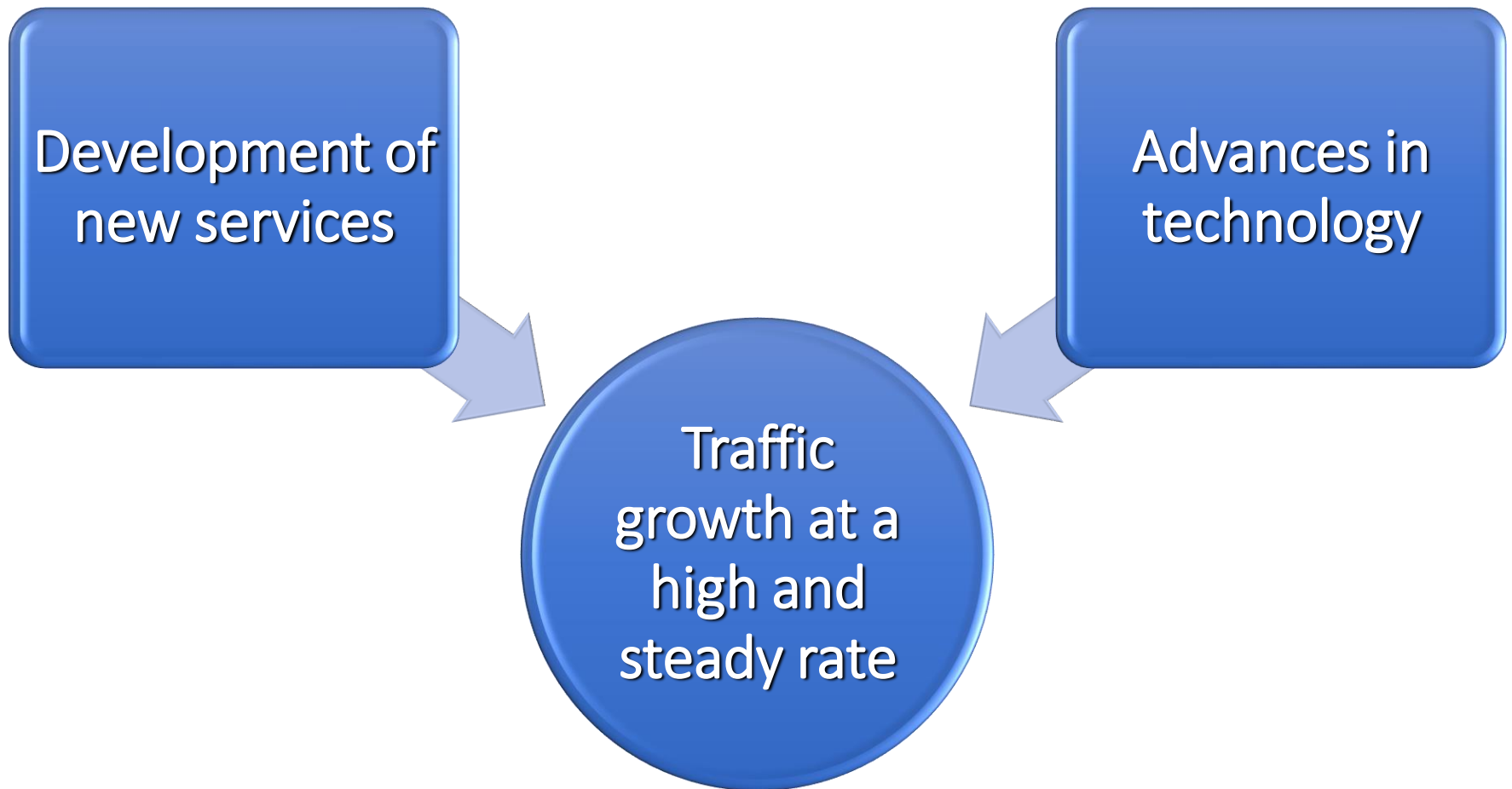
Data Communications

Communication:

- Means **sharing information**
 - Local (face to face) or remote (over distance)
- Telecommunication
 - Telephone, telegraph and television
 - Means communication at a distance
 - Tele is Greek for far

Data communication : is the exchange of data between different devices via some form of transmission medium.

Technological Advancement Driving Forces



Notable Trends

Faster and
cheaper, in both
computing and
communication

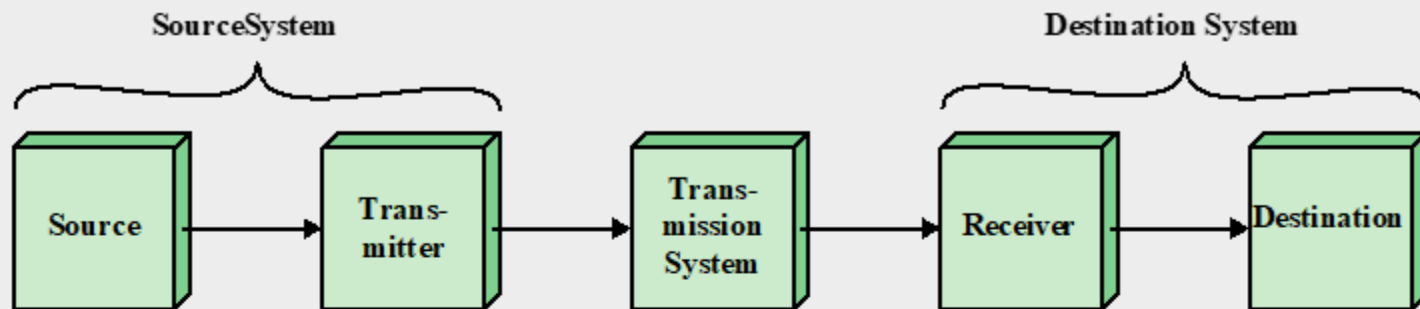
Today's networks
are more
“intelligent”

The Internet

Mobility

Changes in Networking Technology

- **Emergence of high-speed LANs**
- **Corporate WAN needs**
- **Digital electronics**

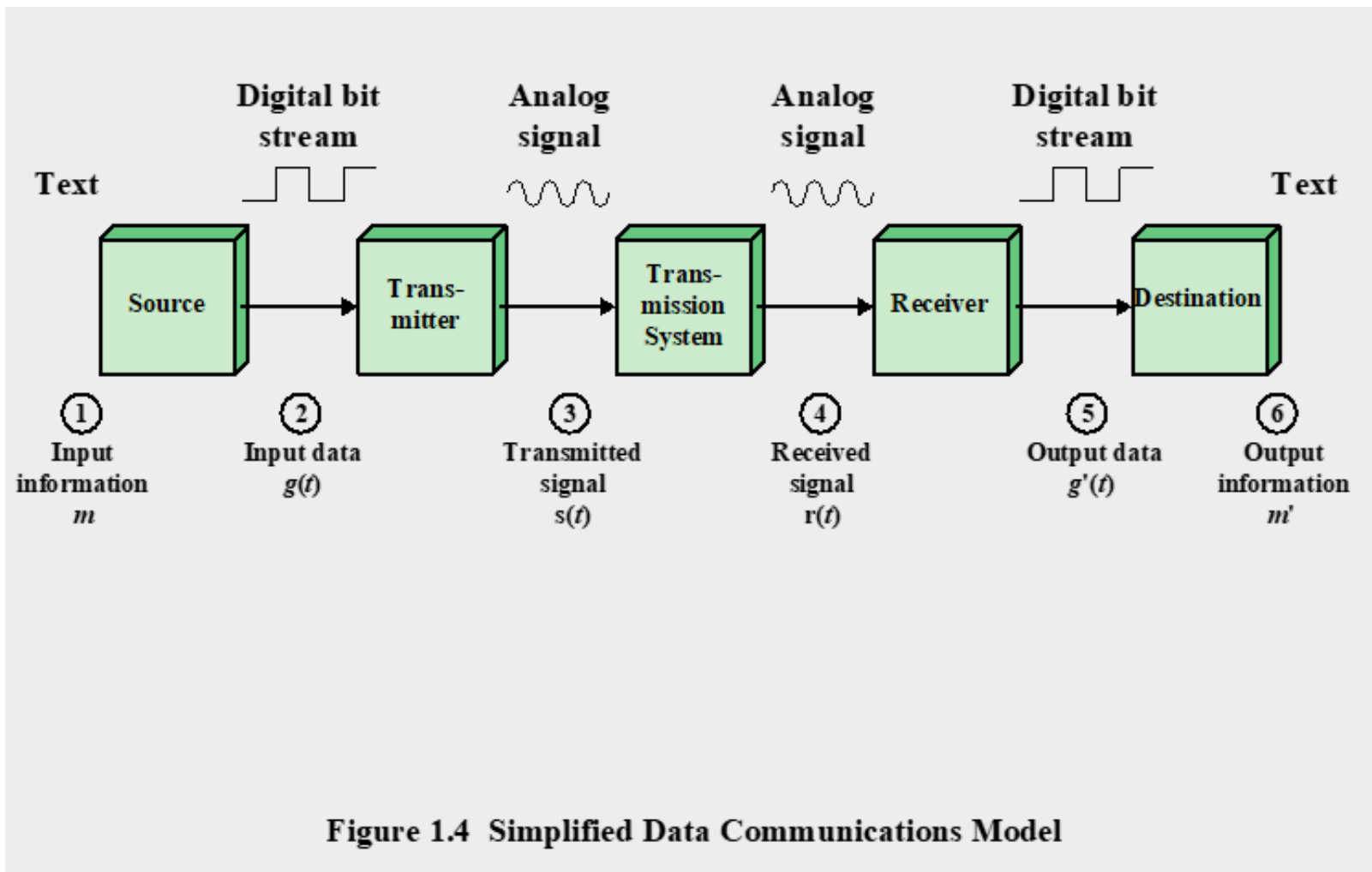


(a) General block diagram



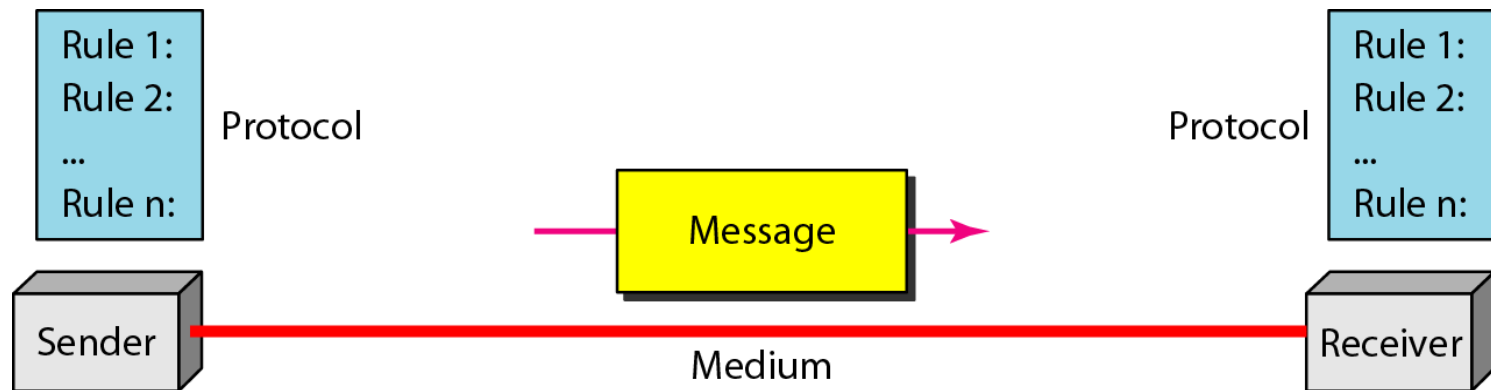
(b) Example

Figure 1.3 Simplified Communications Model



Components

A data communication system is made up of **five** components



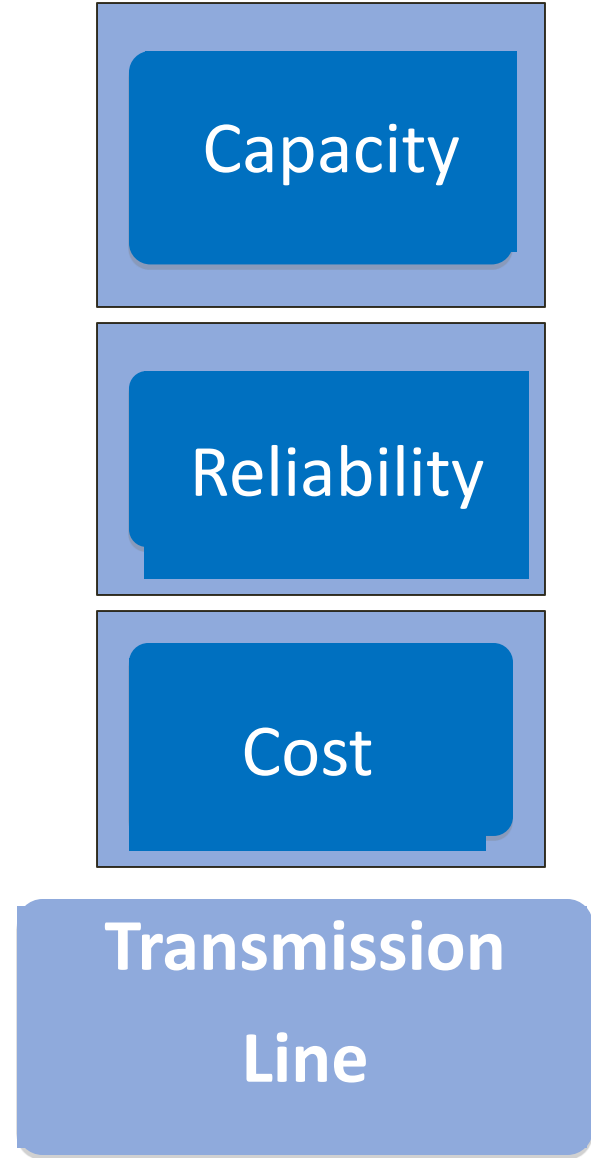
Components

1. **Message**: the information (data) to be communicated
 - Consist of text, numbers, pictures, audio, or video
2. **Sender**: the device that sends the data message
 - Computer, workstation, telephone handset, video camera
3. **Receiver**: the device that receives the message
 - Computer, workstation, telephone handset, television,
4. **Protocol**: a set of rules that govern data communications
5. **Medium**: The physical path by which a message travels from sender to receiver
 - twisted pair, coaxial cable, fiber-optic, radio waves

Transmission Lines

The basic building block of any communications facility is the transmission line

The business manager is concerned with a facility providing the required capacity, with acceptable reliability, at minimum cost



Capacity

Reliability

Cost

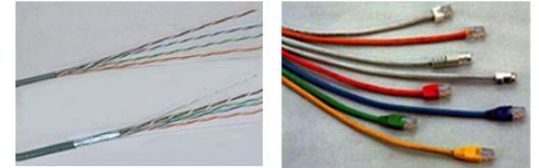
Transmission
Line

Transmission Mediums

Communication across a network is carried through a medium which allows a message to travel from source to destination.

Media Types	Description
Metal wires within cables	Uses electrical impulses
Glass or plastic fibers within cables (fiber-optic cable)	Uses pulses of light.
Wireless transmission	Uses modulation of specific frequencies of electromagnetic waves.

Copper



Fiber-optic



Wireless



Transmission Services

Multiplexing

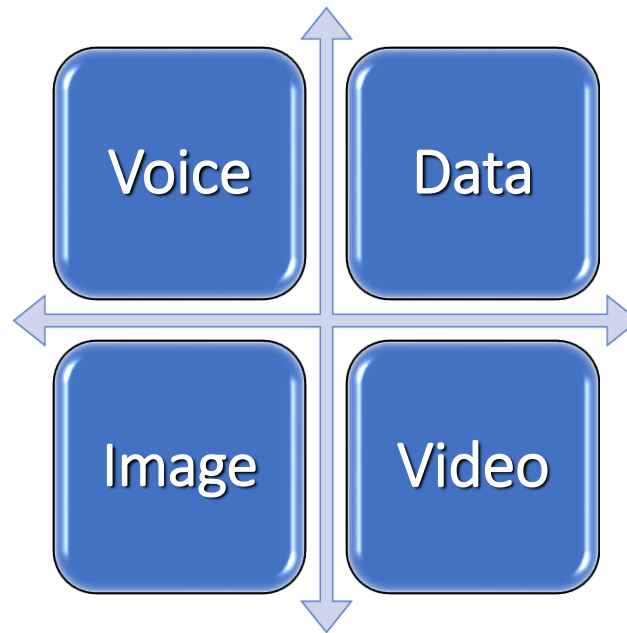
The ability of several devices to share a transmission facility

Compression

Squeezing the data down so that a lower-capacity, cheaper transmission facility can be used

The Converging Network

Converged data networks carry multiple services on one link including data, voice, image, and video



Host Roles

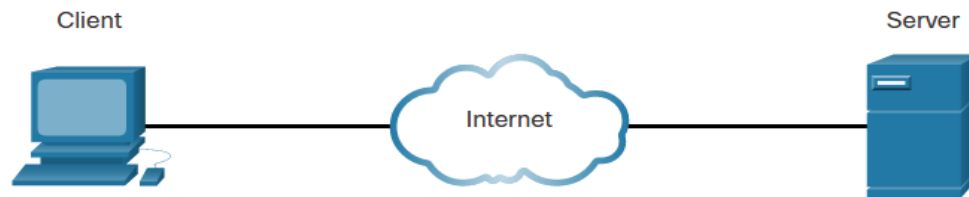
Every computer on a network is called a host or end device.

Servers are computers that provide information to end devices:

- email servers
- web servers
- file server

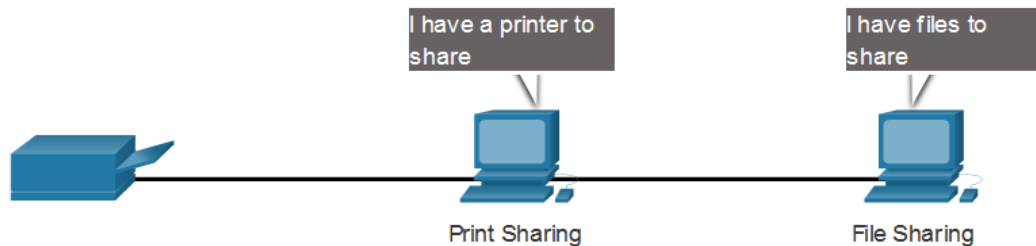
Clients are computers that send requests to the servers to retrieve information:

- web page from a web server
- email from an email server



Peer-to-Peer

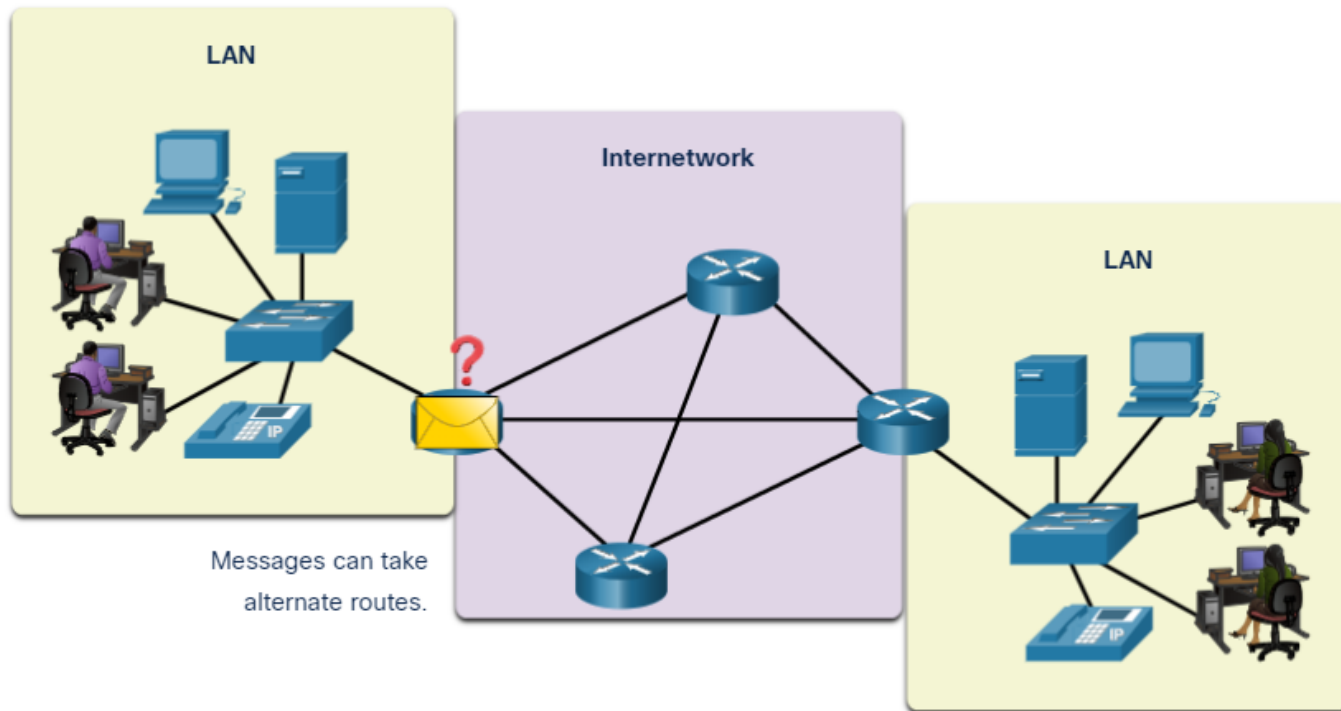
It is possible to have a device be a client and a server in a Peer-to-Peer Network. This type of network design is only recommended for very small networks.



Advantages	Disadvantages
Easy to set up	No centralized administration
Less complex	Not as secure
Lower cost	Not scalable
Used for simple tasks: transferring files and sharing printers	Slower performance

End Devices

An end device is where a message originates from or where it is received. Data originates with an end device, flows through the network, and arrives at an end device.

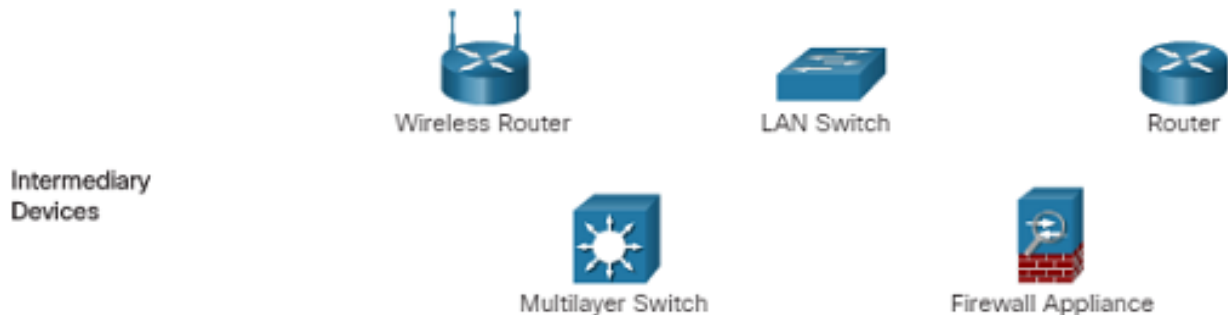


Intermediary Network Devices

An intermediary device interconnects end devices. Examples include switches, wireless access points, routers, and firewalls.

Management of data as it flows through a network is also the role of an intermediary device, including:

- Regenerate and retransmit data signals.
- Maintain information about what pathways exist in the network.
- Notify other devices of errors and communication failures.



Network Representations

End Devices



Intermediary Devices



Network Media



Wireless Media



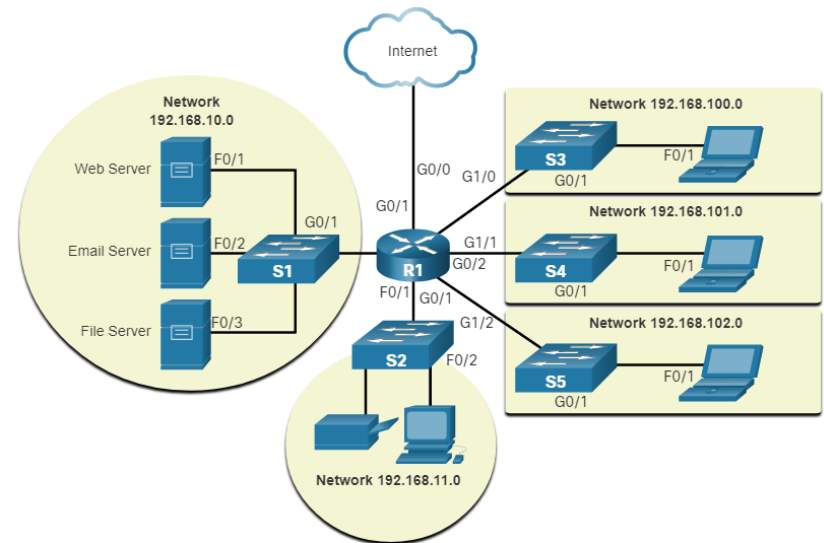
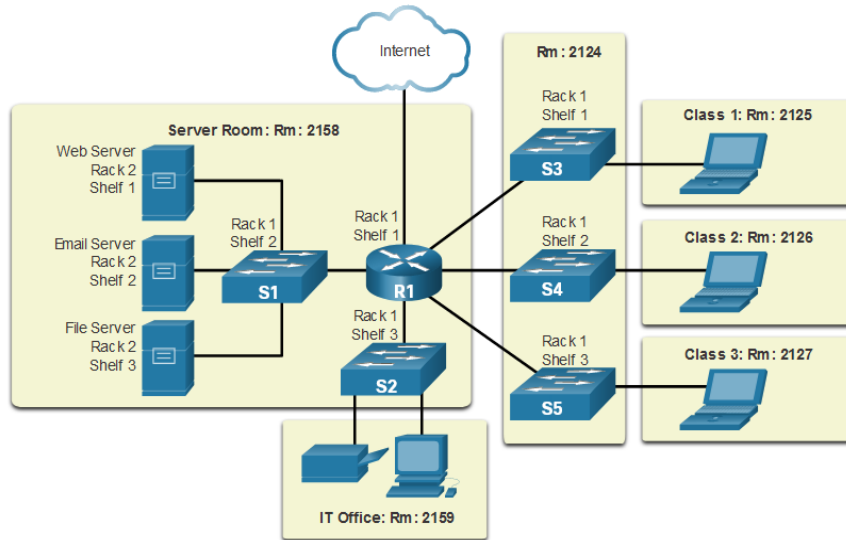
LAN Media



WAN Media

Topology Diagrams

Physical topology diagrams illustrate the physical location of intermediary devices and cable installation.



Logical topology diagrams illustrate devices, ports, and the addressing scheme of the network.

Common Types of Networks



Small Home



SOHO



Medium/Large



World Wide

- Small Home Networks – connect a few computers to each other and the Internet
- Small Office/Home Office – enables computer within a home or remote office to connect to a corporate network
- Medium to Large Networks – many locations with hundreds or thousands of interconnected computers
- World Wide Networks – connects hundreds of millions of computers world-wide – such as the internet

Categories of Networks

- Local Area Network (LAN)
- Metropolitan Area Network (MAN)
- Wide Area Network (WAN)

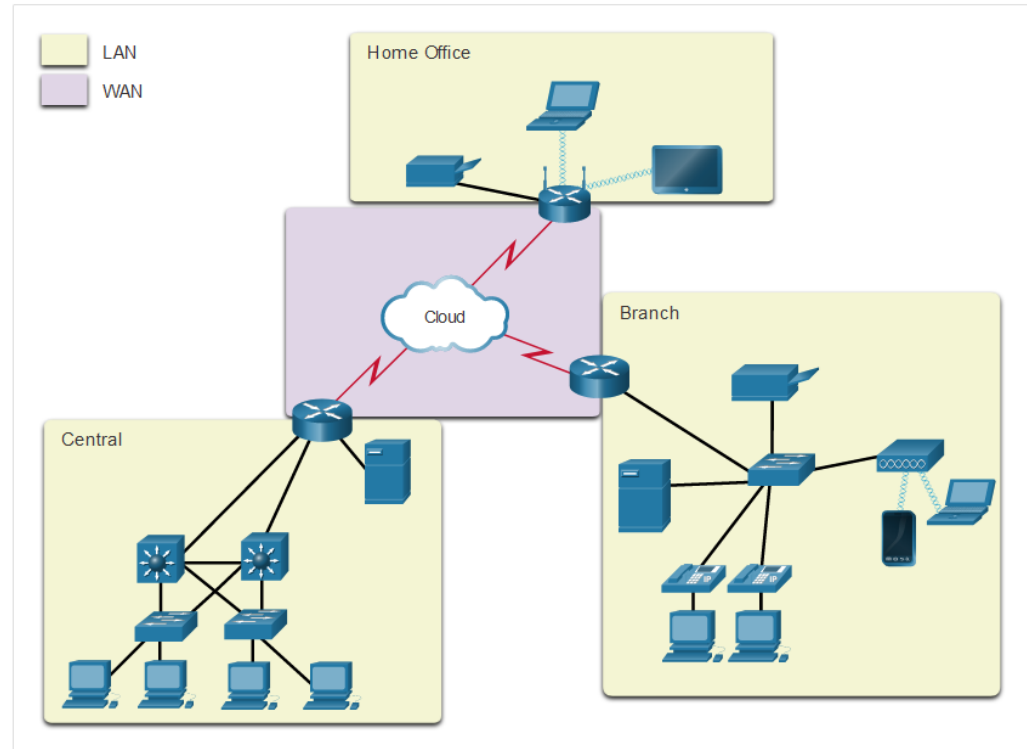
LANs and WANs

Network infrastructures vary greatly in terms of:

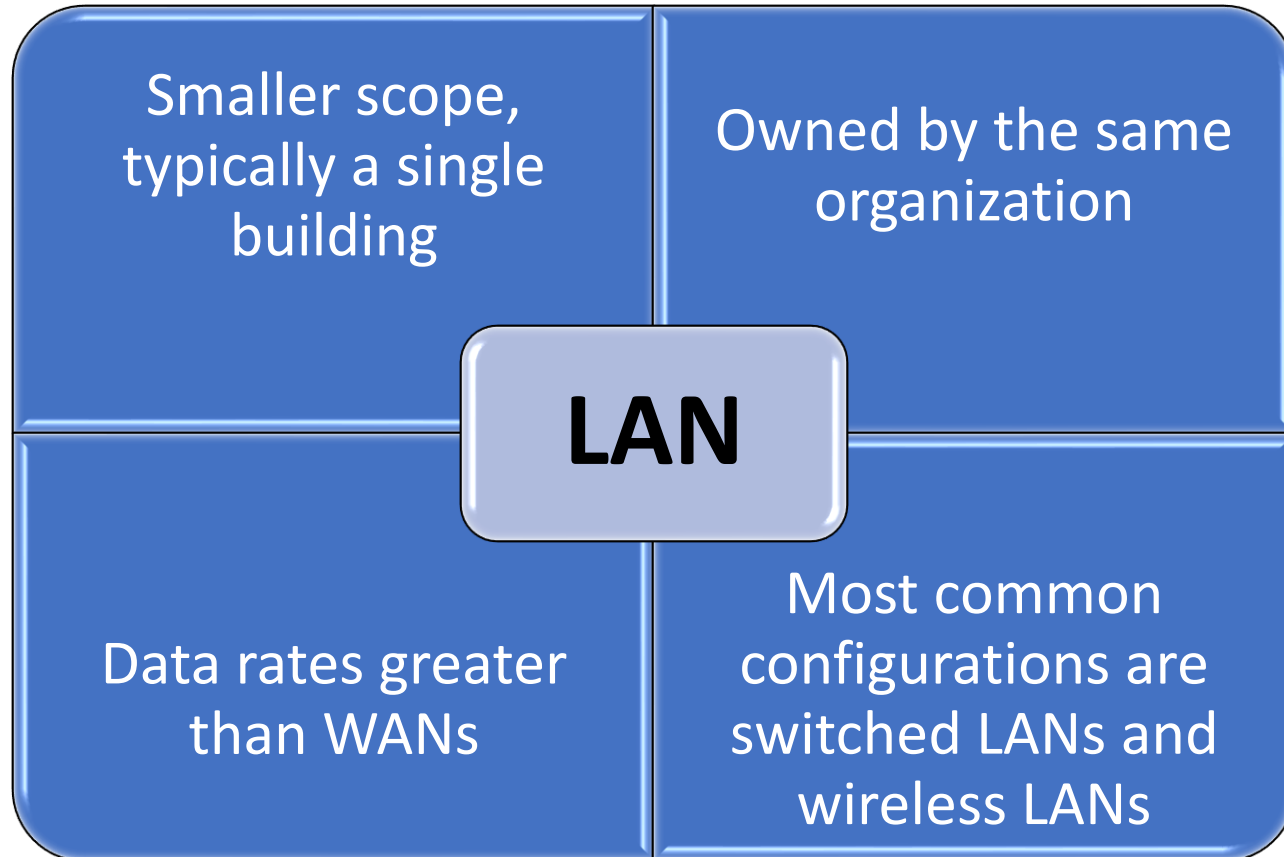
- Size of the area covered
- Number of users connected
- Number and types of services available
- Area of responsibility

Two most common types of networks:

- Local Area Network (LAN)
- Wide Area Network (WAN).



Local Area Networks (LAN)

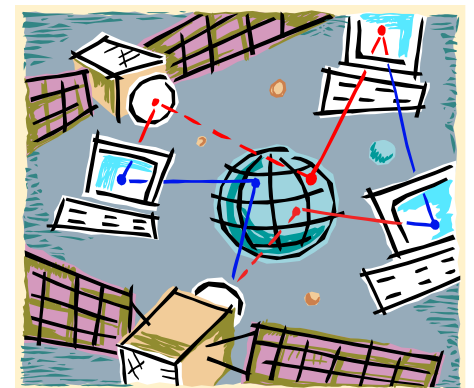


Metropolitan Area Network (MAN)

- middle between LAN and WAN
- private or public network
- high speed
- large area

Wide Area Networks (WANs)

- Span a large geographical area
- Rely in part on common carrier circuits
- Typically consist of a number of interconnected switching nodes

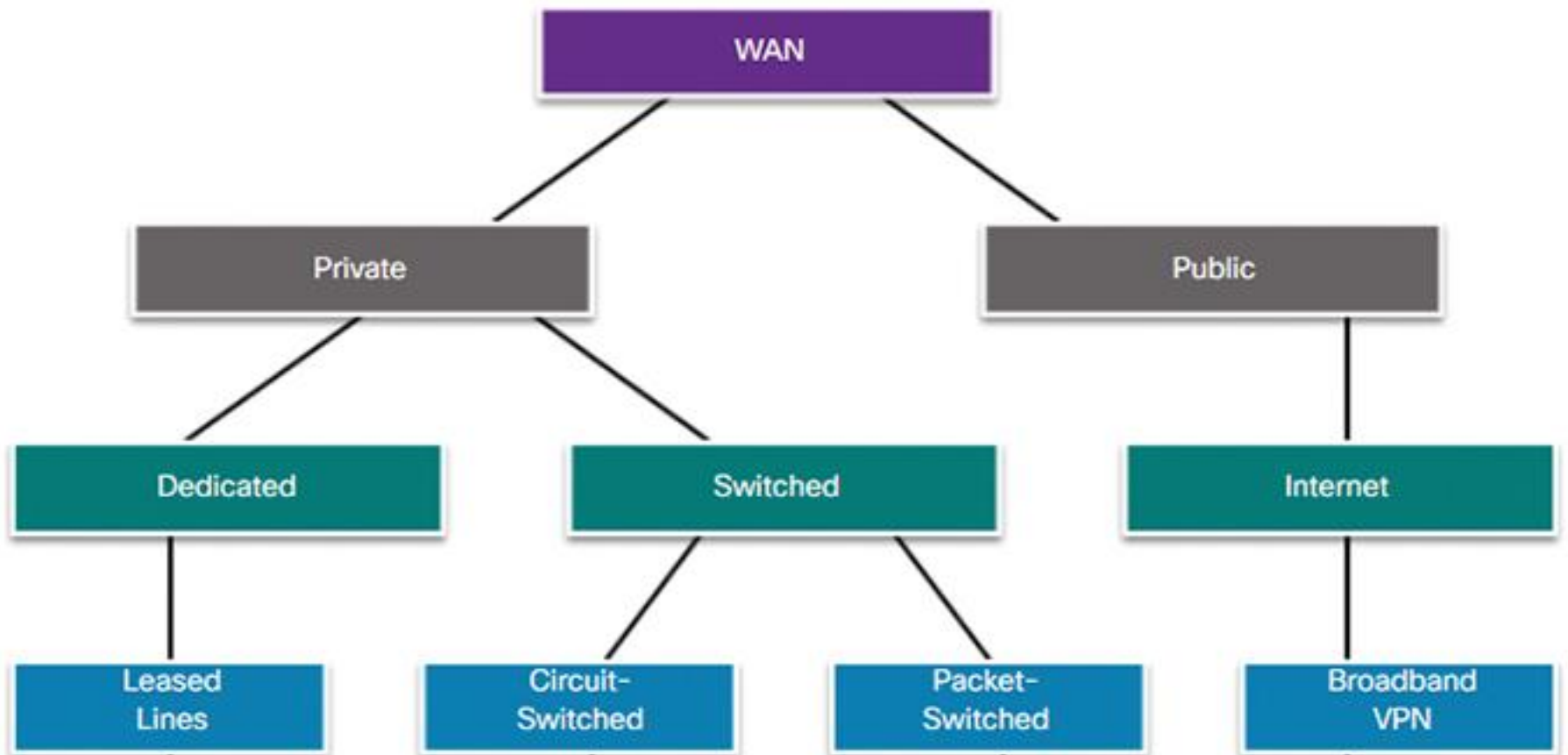


Wide Area Networks

Alternative technologies used include:

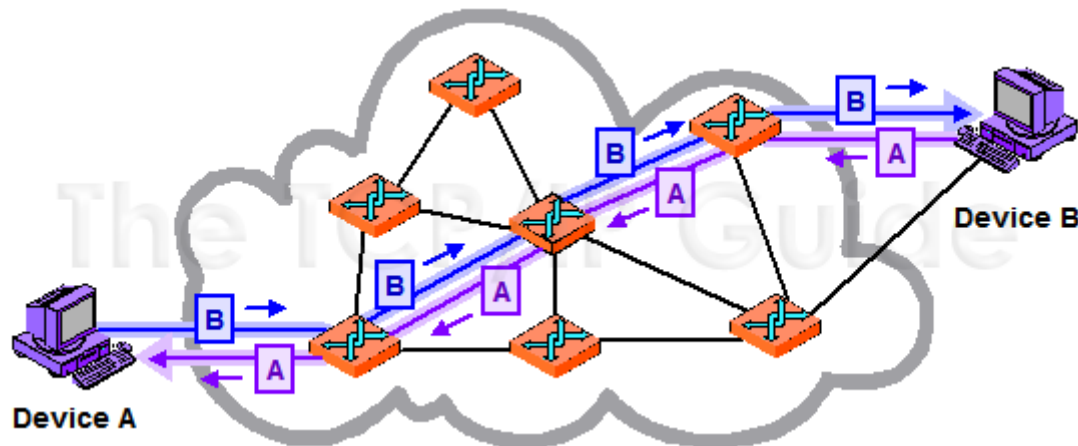
- Circuit switching
- Packet switching
- Frame relay
- Asynchronous Transfer Mode (ATM)





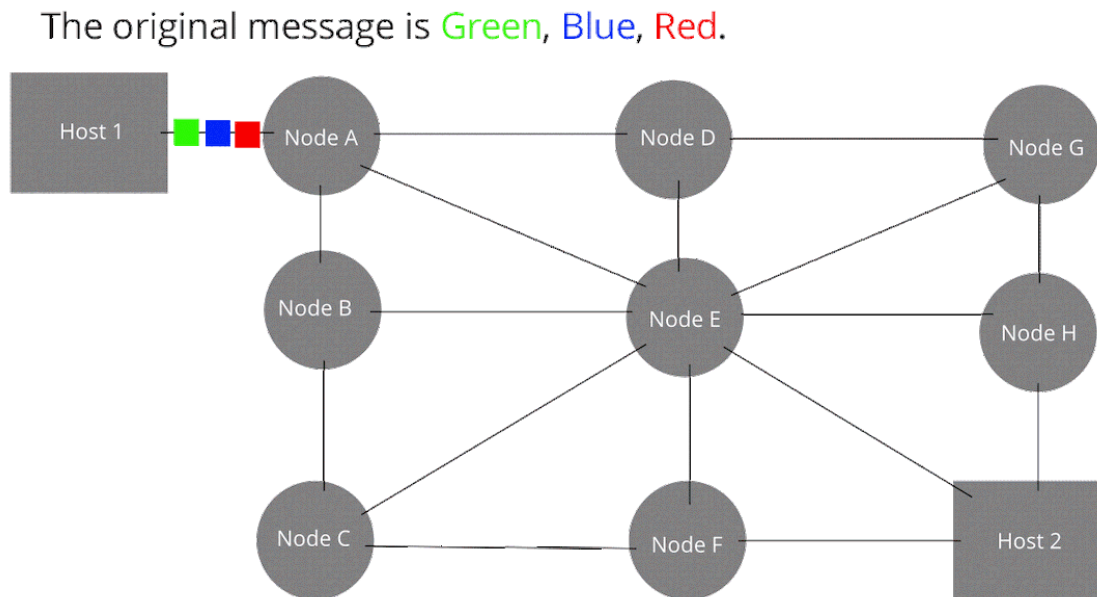
Circuit Switching

- Uses a dedicated communications path
- Connected sequence of physical links between nodes
- The most common example of circuit switching is the telephone network

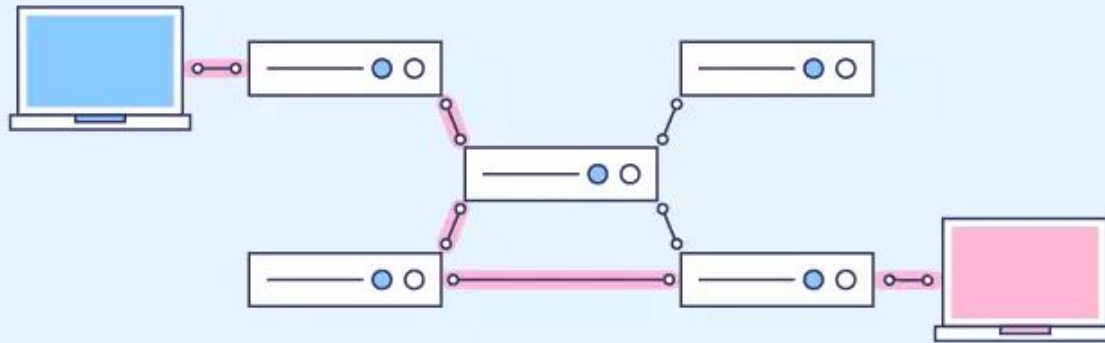


Packet Switching

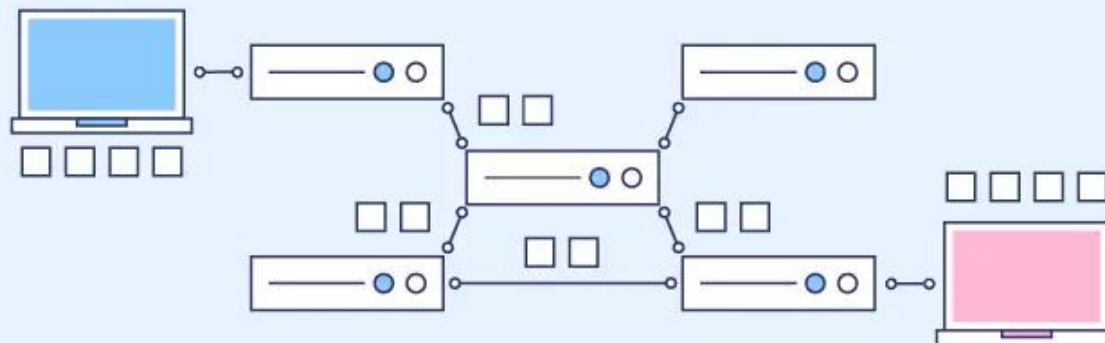
- Data are sent out in a sequence of small chunks called packets
- Packets are passed from node to node along a path leading from source to destination



CIRCUIT SWITCHING



PACKET SWITCHING



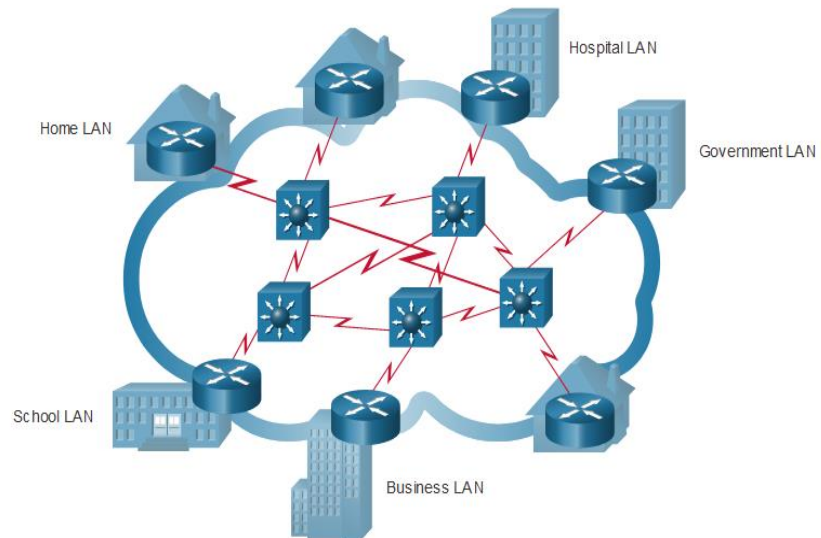
The Internet

The internet is a worldwide collection of interconnected LANs and WANs (Network of networks)

- LANs are connected to each other using WANs.
- WANs may use copper wires, fiber optic cables, and wireless transmissions.

The internet is not owned by any individual or group. The following groups were developed to help maintain structure on the internet:

- IETF
- ICANN
- IAB



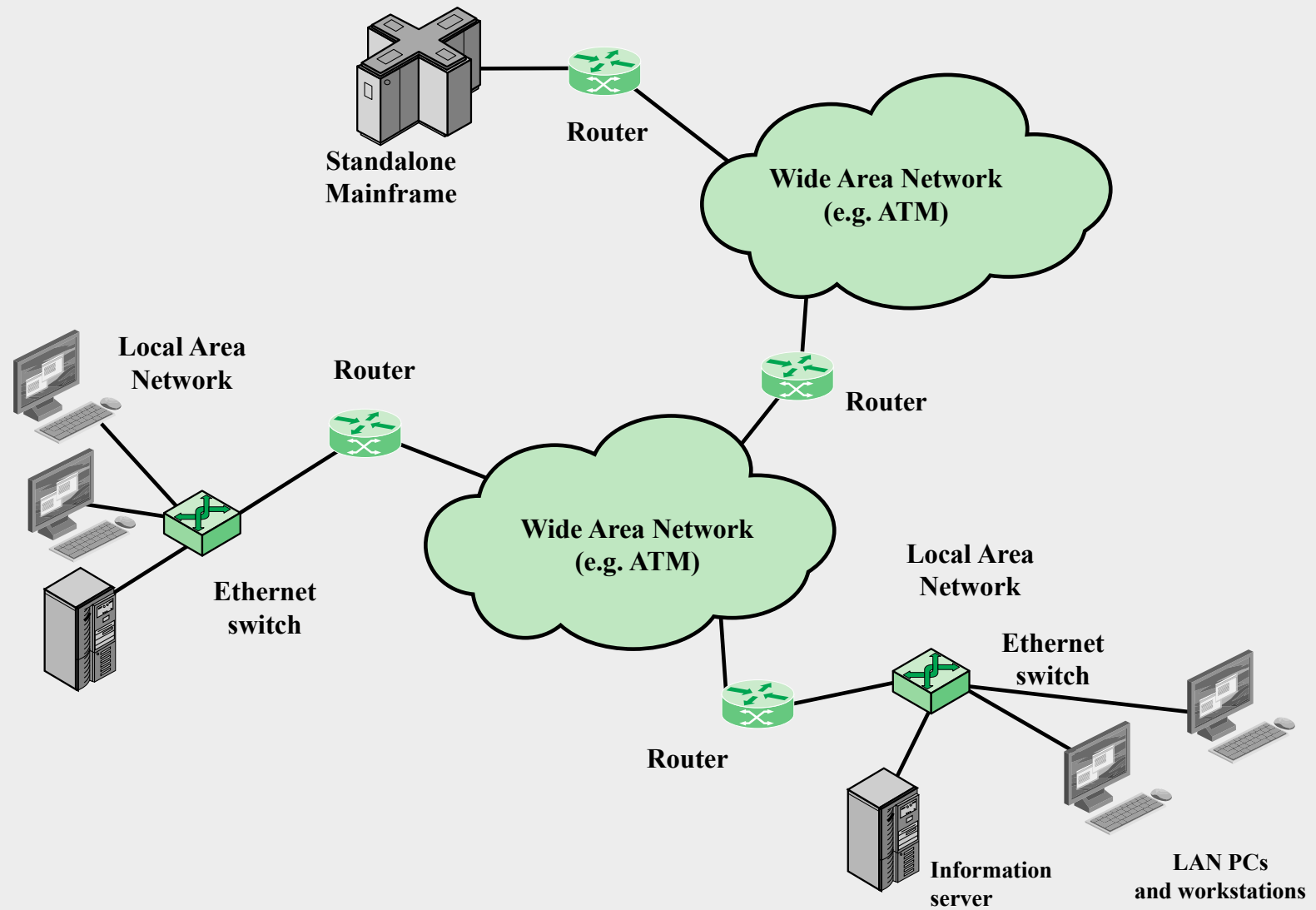


Figure 1.5 Key Elements of the Internet

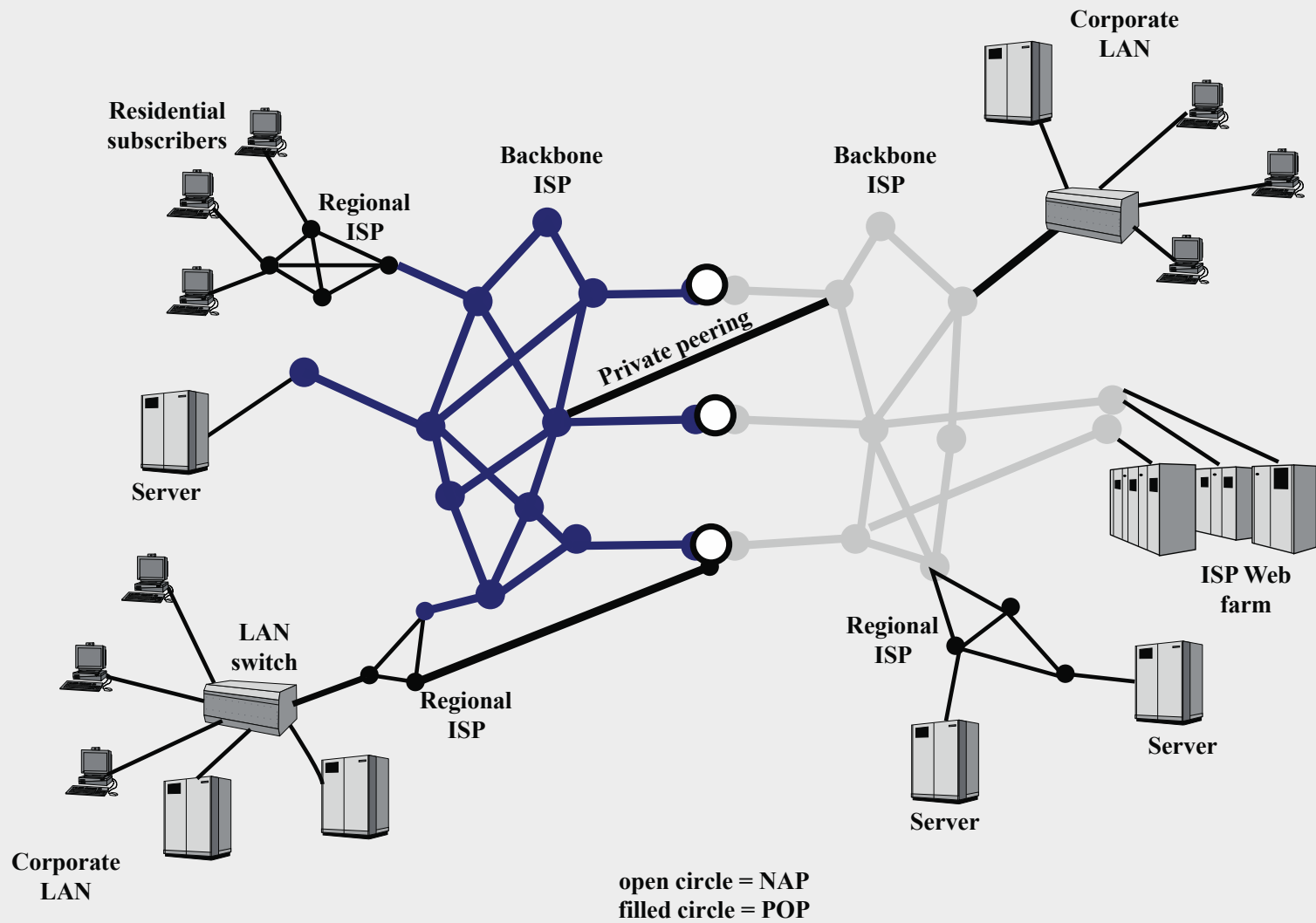
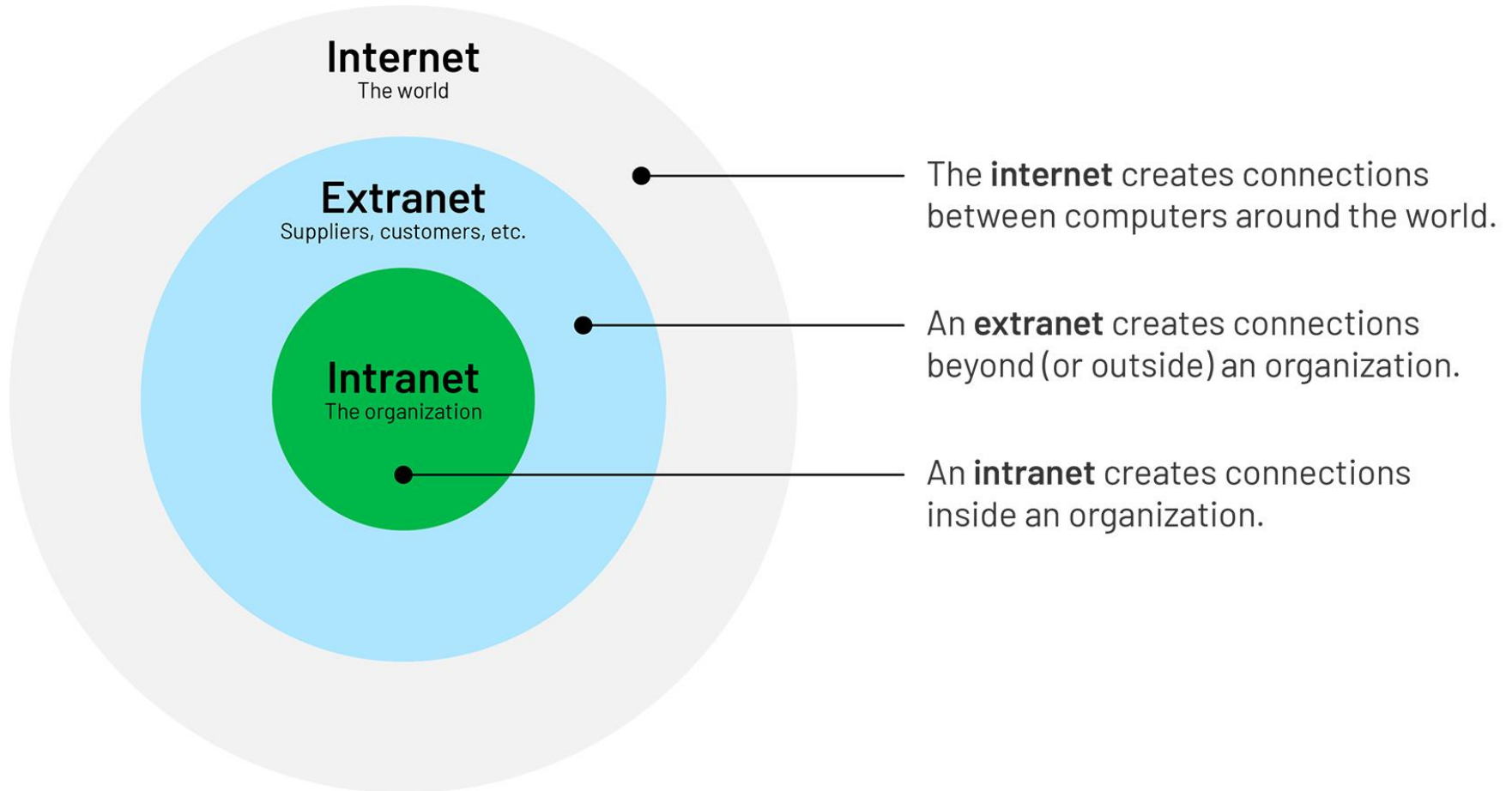


Figure 1.6 Simplified View of Portion of Internet

Internet vs. Intranet vs. Extranet

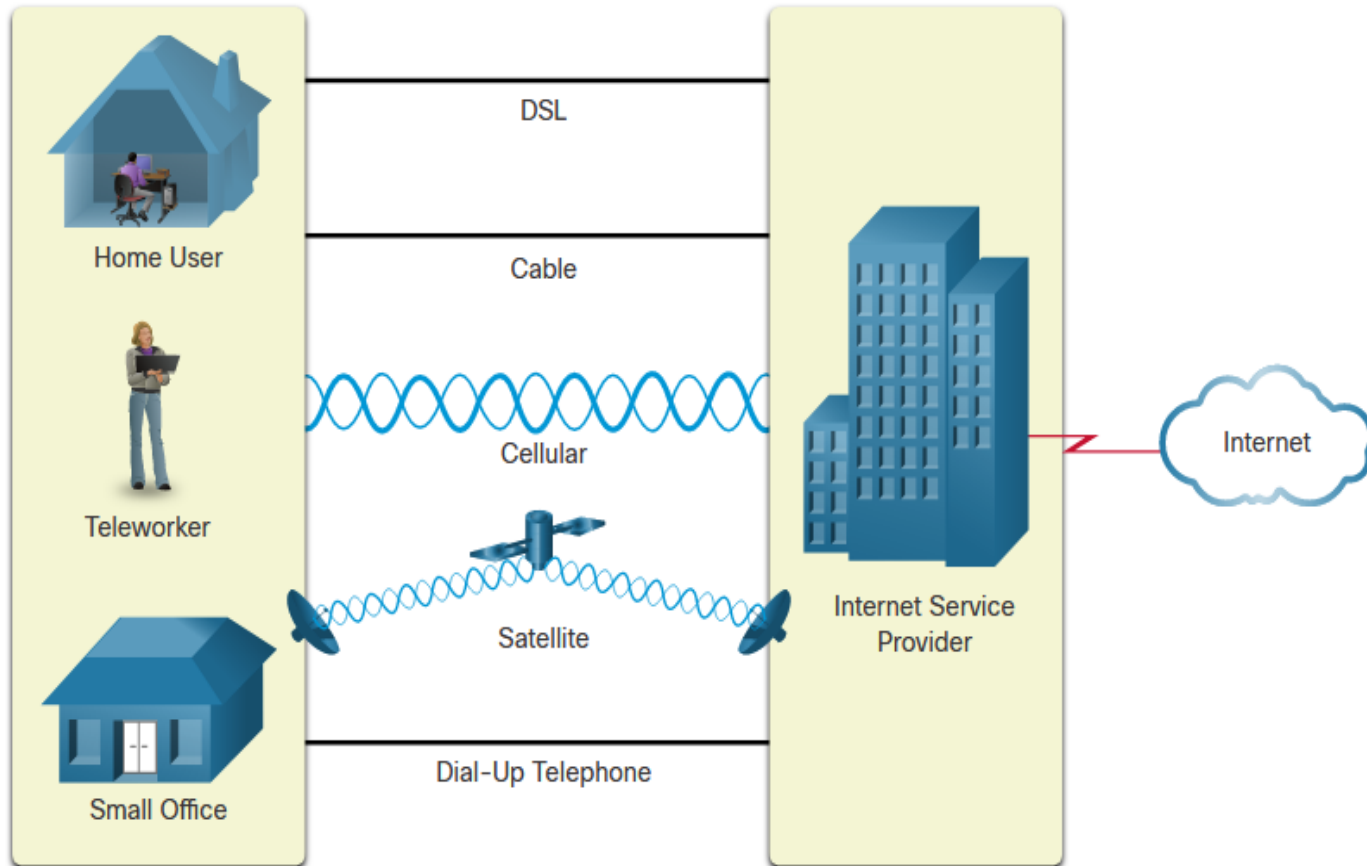


Internet Access Technologies

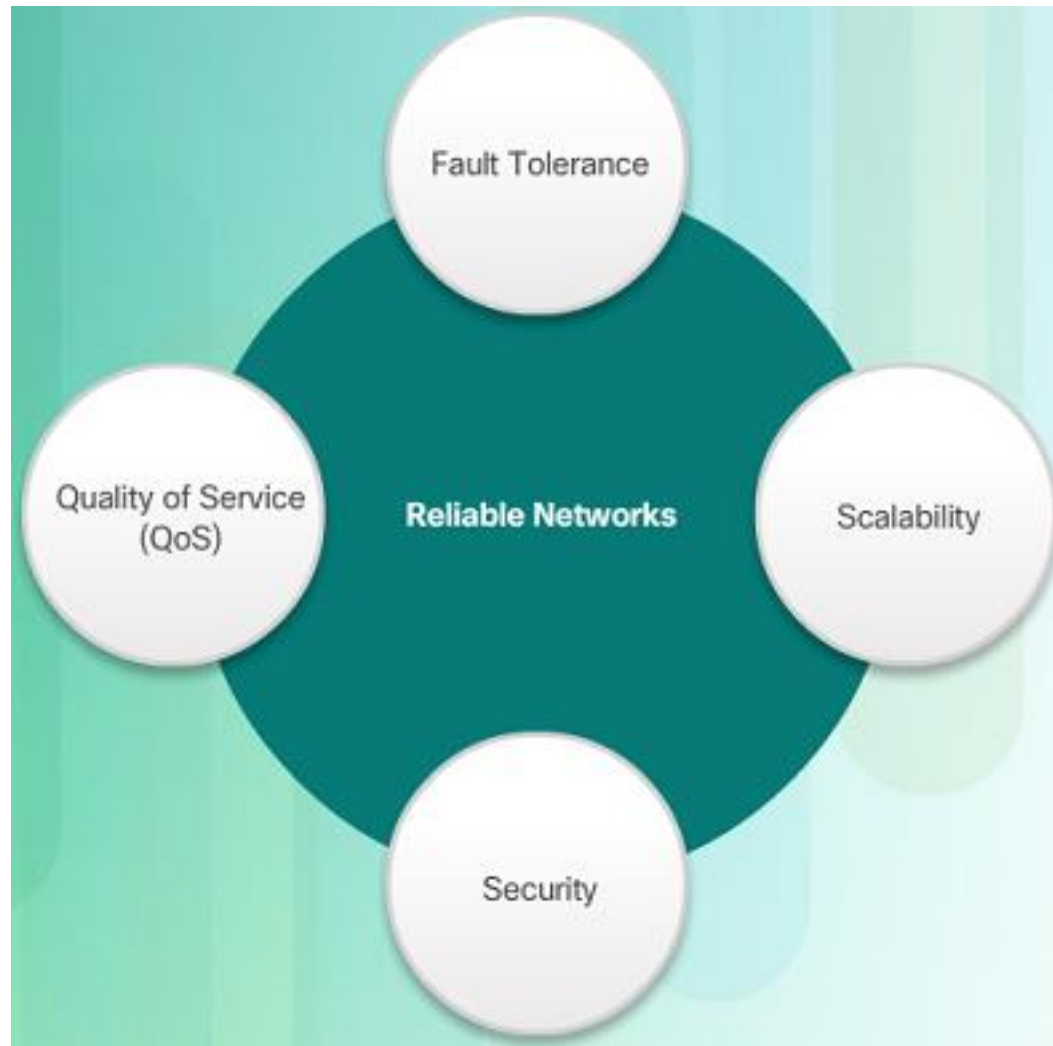
There are many ways to connect users and organizations to the internet:

- Popular services for home users and small offices include broadband cable, broadband digital subscriber line (DSL), wireless WANs, and mobile services.
- Organizations need faster connections to support IP phones, video conferencing and data center storage.

Internet Connections

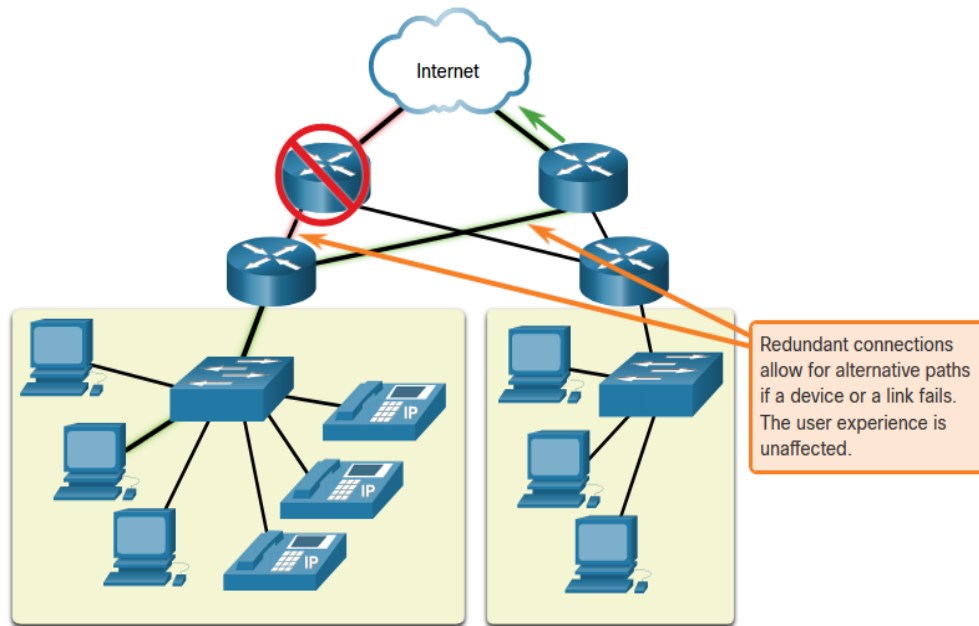


Network Architecture



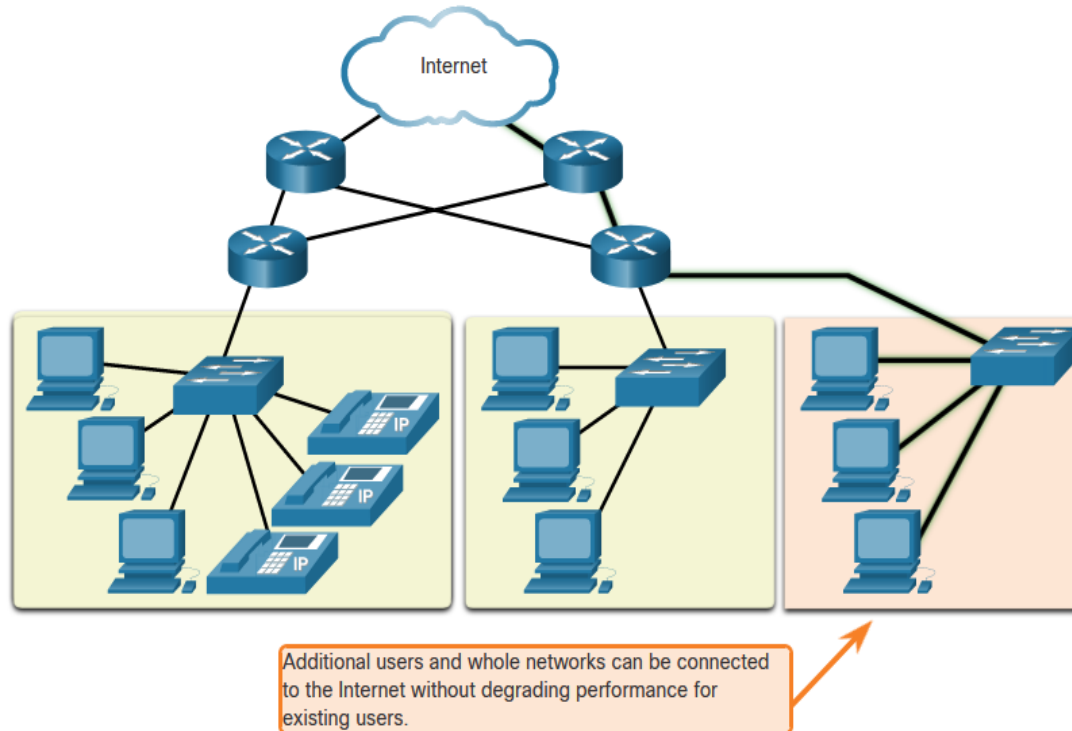
Fault Tolerance

A fault tolerant network limits the impact of a failure by limiting the number of affected devices. Multiple paths are required for fault tolerance.



Scalability

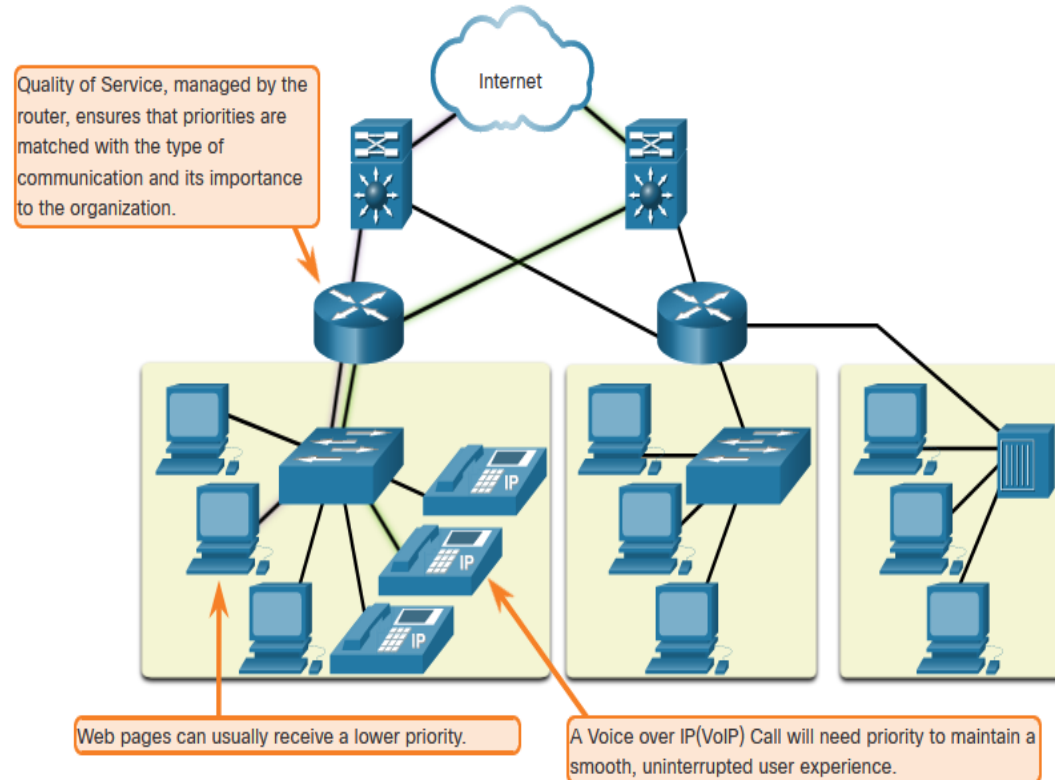
A scalable network can expand quickly and easily to support new users and applications without impacting the performance of services to existing users.



Quality of Service

Voice and live video transmissions require higher expectations for those services being delivered.

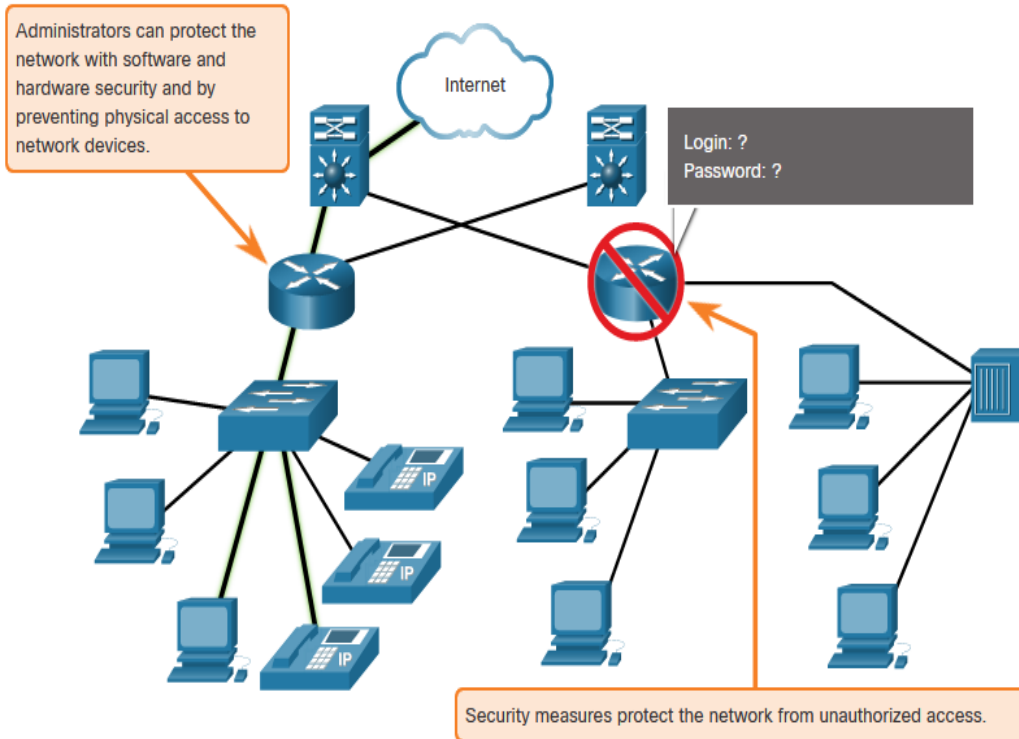
- Quality of Service (QoS) is the primary mechanism used to ensure reliable delivery of content for all users.



Network Security

There are two main types of network security that must be addressed:

- Network infrastructure security
 - Physical security of network devices
 - Preventing unauthorized access to the devices
- Information Security
 - Protection of the information or data transmitted over the network



Network Trends

The role of the network must adjust and continually transform in order to be able to keep up with new technologies and end user devices as they constantly come to the market.

Several new networking trends that effect organizations and consumers:

- Bring Your Own Device (BYOD)
- Online collaboration
- Video communications
- Cloud computing

Bring Your Own Device

Bring Your Own Device (BYOD) allows users to use their own devices giving them more opportunities and greater flexibility.

BYOD allows end users to have the freedom to use personal tools to access information and communicate using their:

- Laptops
- Netbooks
- Tablets
- Smartphones
- E-readers

BYOD means any device, with any ownership, used anywhere.

Online Collaboration

- Collaborate and work with others over the network on joint projects.
- Collaboration is a very high priority for businesses and in education.
- Collaboration tools including Cisco WebEx gives users a way to instantly connect and interact.



Video Communication

- Video calls are made to anyone, regardless of where they are located.
- Video conferencing is a powerful tool for communicating with others.
- Video is becoming a critical requirement for effective collaboration.
- Cisco TelePresence powers is one way of working where everyone, everywhere.

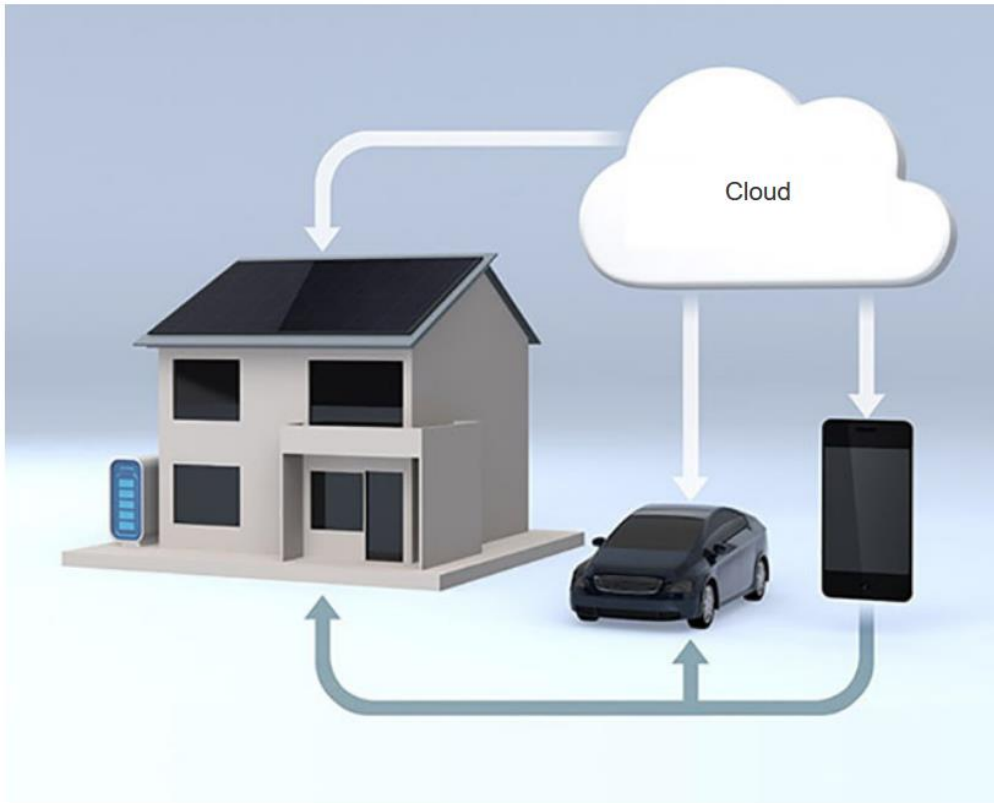
Cloud Computing

Cloud computing allows us to store personal files or backup our data on servers over the internet.

- Applications can also be accessed using the Cloud.
- Allows businesses to deliver to any device anywhere in the world.

Cloud computing is made possible by data centers.

Technology Trends in the Home



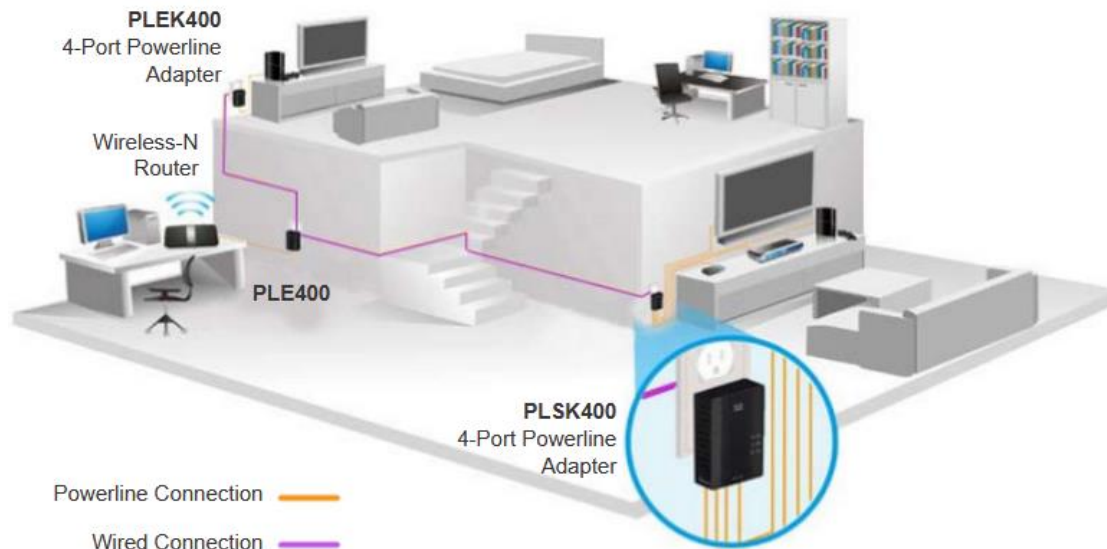
Smart home technology is a growing trend that allows technology to be integrated into every-day appliances which allows them to interconnect with other devices.

Ovens might know what time to cook a meal for you by communicating with your calendar on what time you are scheduled to be home.

Smart home technology is currently being developed for all rooms within a house.

Powerline Networking

- Powerline networking can allow devices to connect to a LAN where data network cables or wireless communications are not a viable option.
- Using a standard powerline adapter, devices can connect to the LAN wherever there is an electrical outlet by sending data on certain frequencies.
- Powerline networking is especially useful when wireless access points cannot reach all the devices in the home.



Wireless Broadband

In addition to DSL and cable, wireless is another option used to connect homes and small businesses to the internet.

- Wireless Internet Service Provider (WISP) is an ISP that connects subscribers to designated access points or hotspots.
- Wireless broadband is another solution for the home and small businesses.

